Osptek Display

3.48" TFT LCD With Touch Panel

Model No:

YDP348BT001-V1

Revision Record

Rev No.	Date	Description
V.0	2026-07-15	First Release

CONTENTS

Revision Record

Contents

ITEM	PAGE
1. General Specifications	4
2. Pin Assignments	5
3. Electrical Specification	6-8
3.1 Absolute Maximum Ratings	6
3.2 Typical Operation Conditions	7
3.3 3.3 Backlight Circuit Characteristics	8
4. Optical Specification	9-10
4.1 LCM Optical Characteristics	9
4.2 Measurement system	10
5. Mechanical Drawing	11
6. Reliability Test Items	12
7. Packing and Storage Specification (Reference only)	13-14
8. Announcements	15

1.General Specifications

No.	Item 项目	Specification 规格	Unit 单位	Remark
1	LCD Size 液晶面板尺寸	3.48	inch	-
2	Panel Type 面板类型	inch	-	-
3	Resolution 分辨率	172*RGBx 640	Pixel	-
4	Display Mode 显示模式	Normally Black	-	-
5	Number of Colors 颜色数量	16.7M	-	-
6	Viewing Direction 使用视角	ALL	-	Note1
7	NTSC 色彩饱和度	70%	-	Typ.
8	Contrast Ratio 对比度	1200	-	
9	Luminance 亮度	300	cd/m2	Typ.
10	Module Size 模组尺寸	28.99 (H)x 92.89 (V)x 2.98 D)	mm	Note1
11	Panel Active Area 可视区域	22.58 x84	mm	Note1
12	Pixel Pitch 像素尺寸	131.25(H)x131.25 (V)	mm	-
13	Pixel Arrangement 像素排列	RGB -stripe		-
14	Weight 重量		g	-
15	Driver IC 驱动芯片	AXS15231B	-	-
16	Driver IC RAM Size 记忆体	RAM less	bit	-
17	Light Source 背光源	6LED	-	-
18	Interface 接口方式	4SPI/QSPI	-	-
19	Operating Temperature 工作温度	-20~+60	℃	-
20	Storage Temperature 存储温度	-30~+70	℃	-

Note 1: Please refer to the mechanical drawing；

注 1：请参照模组图；

2.Pin Assignments

Pin No. Pin 序号	Symbol 符号	Function 功能描述
1	RSTN	Chip reset signal 复位
2	SCL	Serial interface clock 串行时钟引脚。
3	RS	SPI 4wire 8bits：command or parameter selection in spi 4wire 8bits.
4	CS	Chip select pin 片选信号。
5	VDD2.8V	Analog Power Supply for LCM 系统输入电压
6	VDDI2.8V	Power Supply
7	DIN_SDA_DUAL	SPI ：串行数据 /QSPI:QSPI interface data bus
8	GND	Ground 接地。
9	LEDK	Backlight cathode 背光负极输入端。
10	LEDA	Backlight anode 背光正极输入端
11	GND	Ground 接地。
12	TP INT	Interrupt signal for TP
13	TP SDA	I2C SDA,TYP
14	TP SCL	I2C CLK,TYP.
15	TP RST	The external reset input for TP
16	IM3	IM3=“0”:4SPI; IM3=“1”:QSPI;
17	DIN_SDA	QSPI interface data bus.
18	D2	QSPI interface data bus.
19	D3	QSPI interface data bus.
20	TE	Tearing effect signal is used to synchronize MCU to frame memory writing

3. Electrical Specification

3.1Absolute Maximum Ratings

Item 项目	Symbol	Value	Unit	Remark
Analog Power Supply Voltage 模拟供电电压	VCI	-0.3~+3.3	V	-
Digital Power Supply Voltage 数字电源电压	VDD	-0.3~3.3	V	-
I/O Power Supply Voltage I/O端口供电电压	IOVCC	-0.3~3.3	V	-

3. Electrical Specification

3.2 Typical Operation Conditions

Item 项目	Symbol	Min.最小	Typ.典型	Max.最大	Unit
Analog Supply Voltage 模拟供电电压	VCI	2.7	2.8	3.3	V
Digital Supply Voltage 数字电源电压	VDD	2.7	2.8	3.3	V
I/O Supply Voltage 接口电压	IOVCC	1.65	1.8/2.8	3.3	V
Input High Voltage 输入高电平	V_{IH}	0.8*IOVCC	-	IOVCC	V
Input Low Voltage 输入低电平	V_{IL}	0	-	0.2*IOVCC	V
Output High Voltage 输出高电平	V_{OH}	0.8*IOVCC	-	-	V
Output Low Voltage 输出低电平	V_{OL}	-	-	0.2*IOVCC	V

osptek®

3.3 Backlight Circuit Characteristics

Item	Symbol	Min.	Typ.	Max.	Unit
LED Current 背光电流	I_B		20	-	mA
LED Voltage 背光电压	Vf		18.6		V
Power Consumption 功耗	P_{BL}	-		-	mW

3.4 LCD Current Consumption 液晶面板功耗

Item	Symbol	Typ.	Max.	Unit
Full Mode 正常模式	VCI	-	-	mA
测试条件：VCI=2.8V，IOVCC=2.8V； Interface 驱动类型：行翻转或者列翻转； TN Type=>All Black Pattern. TN型液晶面板=>黑色画面； IPS Type=>All White Pattern. IPS型液晶面板=>白色画面； Temperature：25℃；温度：室温25摄氏度；				
Sleep Mode 休眠模式	VCI	-	-	uA
测试条件：VCI=2.8V，IOVCC=2.8V； DC/DC converter is enabled. Internal oscillator is started and panel scanning is started. 除IC内部晶振和面板扫描外，其他功能都暂停工作； Temperature：25℃；温度：室温25摄氏度；				

4.Optical Specification

4.1 LCM Optical Characteristics

Item		Symbol	Condition	Min.	Typ.	Max.	Unit
Viewing Angle Range 视角	horizontal	Θ3	CR≧10	75	80	-	degree
		Θ9		75	80	-	
	vertical	Θ12		75	80	-	
		Θ6		75	80	-	
Response Time 响应时间		Ton+Toff	θ=Φ=0°	30	35		ms
Contrast Ratio 对比度		CR	θ=Φ=0°	900	1200	-	-
Luminance 亮度		L	θ=Φ=0°	-	300	-	cd/m^2
Color Chromaticity (CIE1931) 色坐标	White	W_x	Normal θ=Φ=0°		0.308		-
		W_y		Typ-0.03	0.337	typ+0.03	
Uniformity 均匀度		U_L	θ=Φ=0°	80	-	-	%
Cross Talk		ct	-	2.0%			-

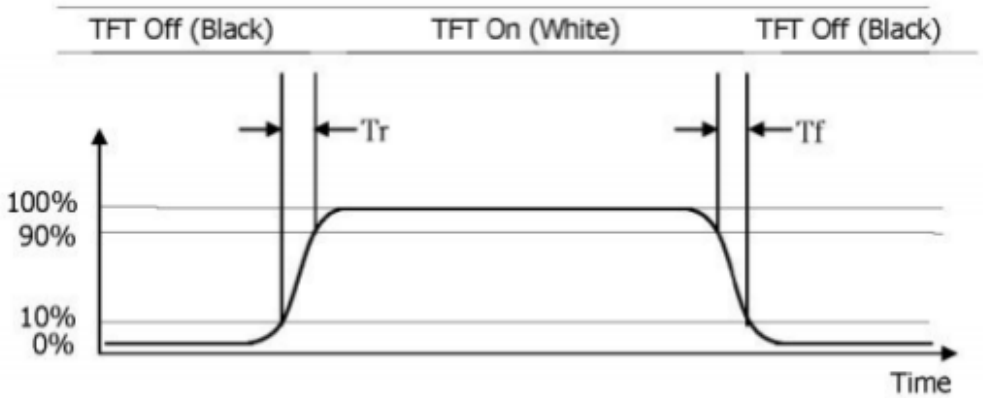
4.2 Measurement system

4.2.1 LCM Viewing Angle

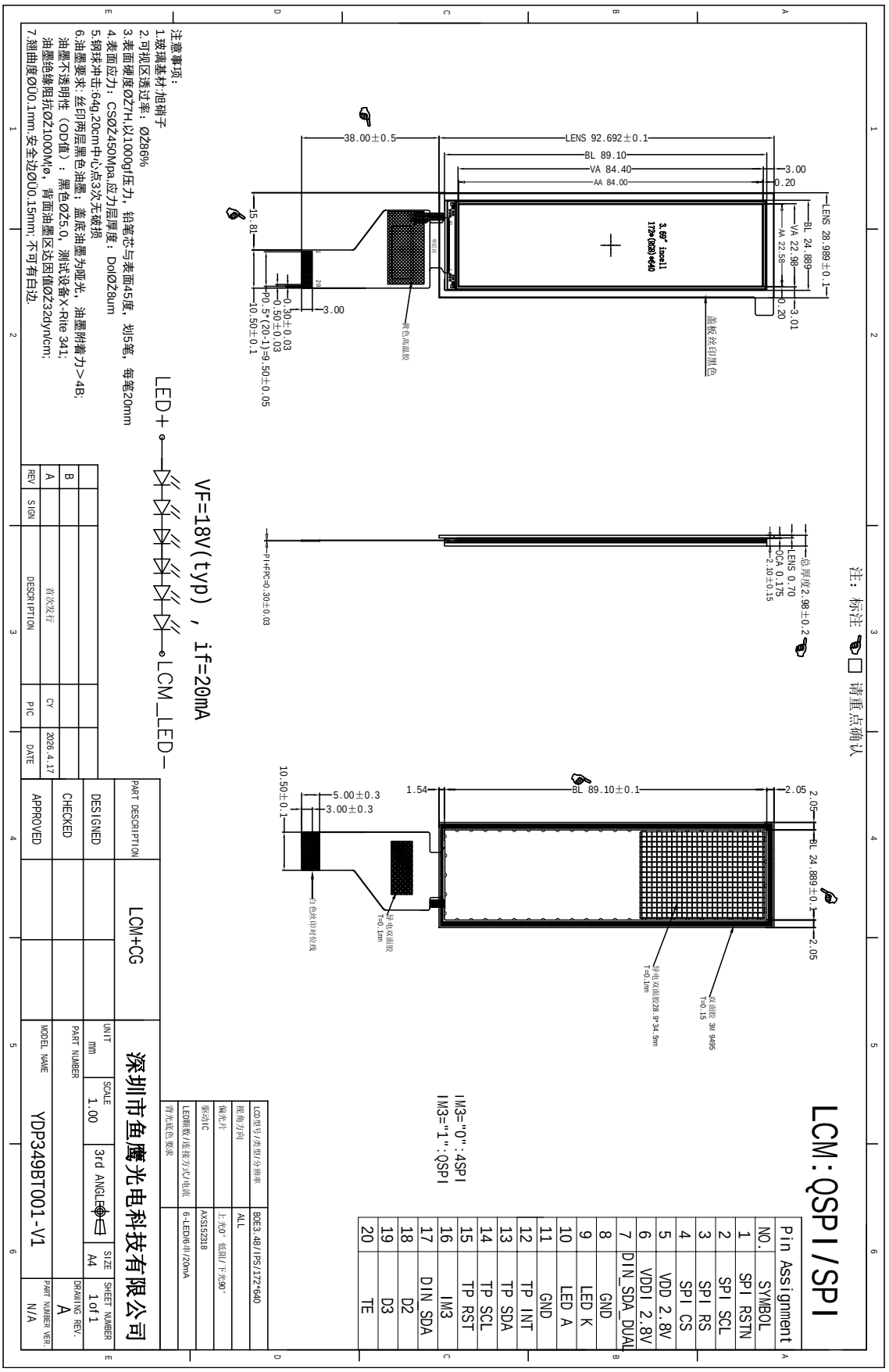
1. Viewing angle is the angle at which the contrast ratio is greater than 10. The viewing are determined for the horizontal or 3, 9 o'clock direction and the vertical or 6, 12 o'clock direction with respect to the optical axis which is normal to the LCD surface.
2. Contrast measurements shall be made at viewing angle of θ= 0。 and at the center of the LCD surface. Luminance shall be measured with all pixels in the view field set first to white, then to the dark (black) state. (See FIGURE 1 shown in Appendix) Luminance Contrast Ratio (CR) is defined mathematically.

$$CR = \frac{\text{Luminance when displaying a white raster}}{\text{Luminance when displaying a black raster}}$$

3. Center Luminance of white is defined as the LCD surface. Luminance shall be measured with all pixels in the view field set first to white. This measurementshall be taken at the locations shown in FIGURE 4 for a total of the measurements per display.
4. The White luminance uniformity on LCD surface is then expressed as : ΔY = Minimum Luminance of 9 Points or 5 points / Maximum Luminance of 9 Points or 5 points(See FIGURE 2).
5. The color chromaticity coordinates specified in Table 5. shall be calculated from the spectral data measured with C light. Measurements shall be made at the center of the panel.
6. The electro-optical response time measurements shall be made as FIGURE 5 shown in Appendix by switching the "data" input signal ON and OFF. The times needed for the luminance to change from 10% to 90% is Td, and 90% to 10% is Tr.



5. Mechanical Drawing



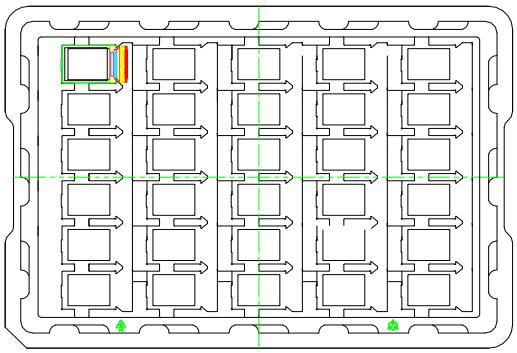
6. Reliability Test Items

Test Item 测试项目	Test Condition 测试条件	Test result determinant gist 实验结果判定
High temperature storage 高温存储	70±3℃，24H；	Inspection after 2~4hours storage at room temperature, the sample shall be free from defects: 试验结束后,已测试的LCD样品必须在室内正常温湿度环境下放置2~4个小时以上才能进行功能和外观检查 ,样品不允许有以下缺陷： 1.Air bubble in the LCD; 模块中有气泡； 2.Non-display; 不显示； 3.Glass crack; 玻璃破碎； 4. The electrical characteristics requirements shall be satisfied. 需要满足模块电气性能。
Low temperature storage 低温存储	-30±3℃，24H；	
High temperature operation 高温运行测试	60±3℃，24H；	
Low temperature operation 低温运行测试	-20±3℃，24H；	
High temperature /humidity 高温高湿	60℃±3℃,90%±3%RH，24H；	
Thermal Shock 冷热冲击	-10℃/0.5h~+60℃/0.5h for a total 24 cycles；	
Vibration Test 振动测试	Frequency 10Hz~55Hz~10Hz Amplitude：1.5mm, X，Y，Z direction for total 1H； (Packing condition)	
ESD test 静电测试	±2KV, Human Body Mode, 150pF/330Ω； ±8KV, Air Mode, 150pF/330Ω；	

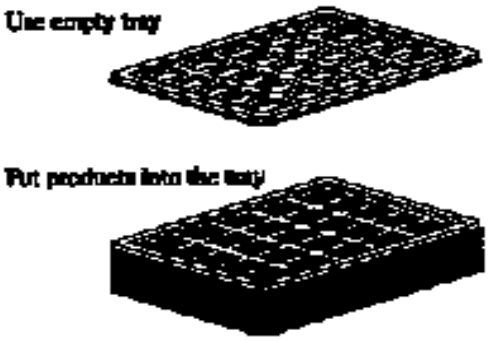
7. Packing and Storage Specification(Reference Only)

7.1 Packing Method

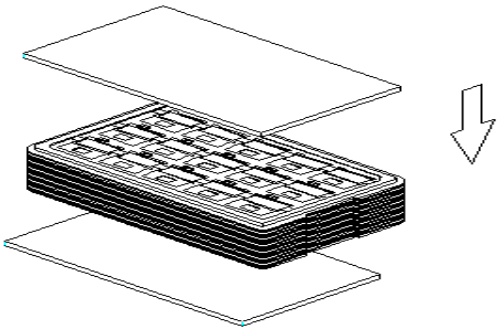
(1)



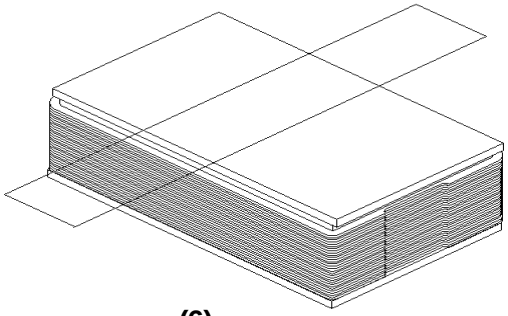
(2)



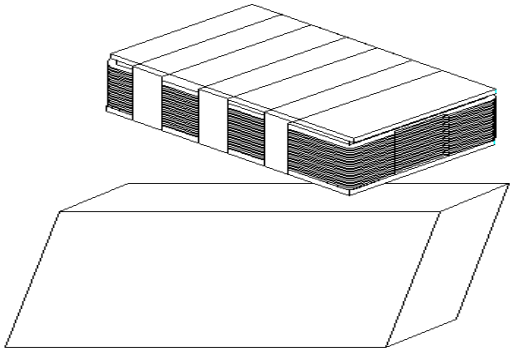
(3)



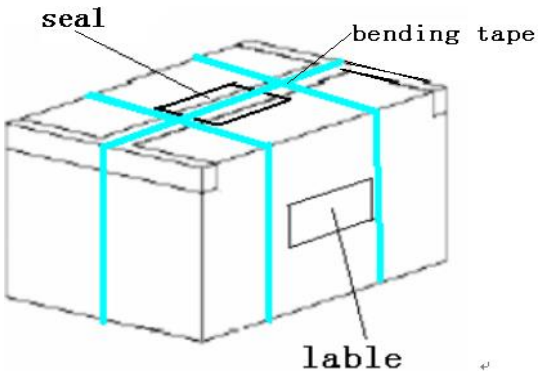
(4)



(5)



(6)



1. Put module into tray cavity. .
2. Tray stacking. .
3. Put 1 foam under the tray stack and 1 foam above. .
4. Fix the cardboard to the tray stack with adhesive tape. .
5. Put the tray stack into carton. .
6. Carton sealing with adhesive tape. .

7.2 Storage Method

1. Store in an ambient temperature of 23°C±5°C, and in a relative humidity of 55%±15%. Don't exceed 12 months and expose to sunlight or fluorescent light.
2. Store in a clean environment, free from dust, active gas, and solvent.
3. Store in antistatic container.

8. Announcements

1. Do not attempt to disassemble or process the LCD module.
2. Do not make extra holes on the printed circuit board, modify its shape or change the positions of components to be attached。
3. Except for soldering the interface, do not make any alterations or modifications with a soldering iron; Ensure welding temperature at 320 ° C to 350 ° C, the welding time control within the 10 s, welding note don't stay too long in the same place to avoid scald FPC.
4. Other matters in not clear before use, please contact our staff to guide.